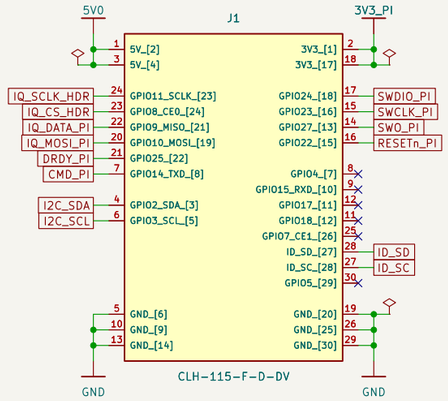
Raspberry Pi header

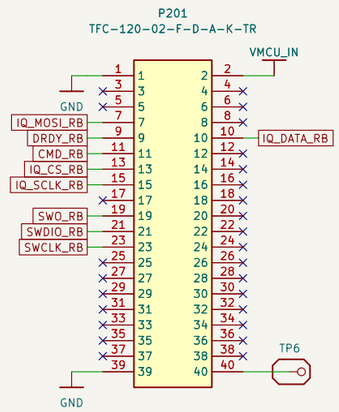
J1 = Samtec CLH-115 on RPi pins 1-30. Pin number = J1 (socket), [n] = RPi header pin.



Pi 3V3 (1, 17) feeds only the ID EEPROM. ID_SD / ID_SC: EEPROM + 3.9k only.
cmdlink: PL011 on GPIO14 (dtoverlay=disable-bt).
Mode A: dtoverlay=spi0-1cs.

P201 — radio board header

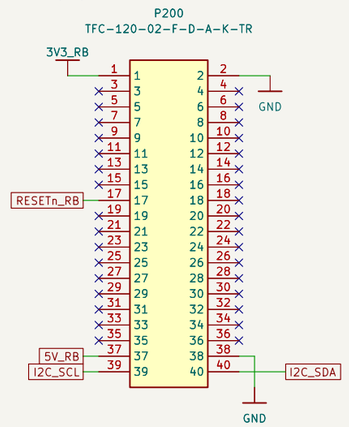
I/Q SPI (EUSART), DRDY, cmdlink, SWD, VMCU [BRD4265B expansion]



P201-40 VRF_IN unused on BRD4265B (on-board DC-DC feeds PAVDD): test pad only.
Pin map = ionos-sdr-mini rev-B3 (validated).

P200 — radio board header

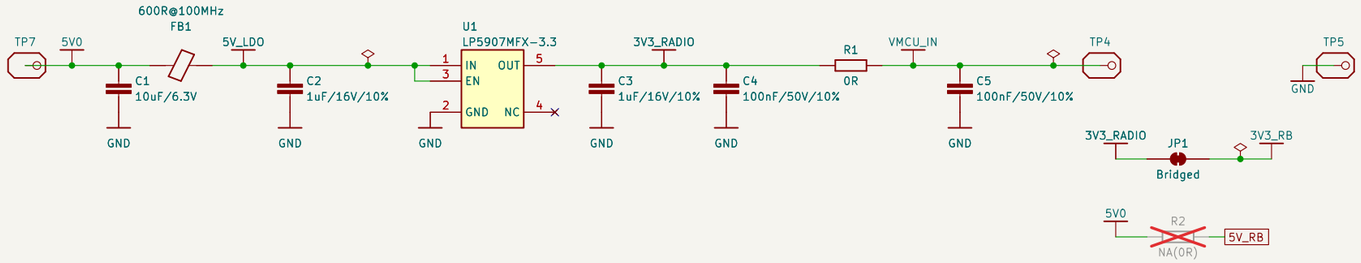
supply, reset, board-ID EEPROM (I2C)



Board-ID EEPROM (M24C02, radio board) on Pi I2C1;
pull-ups 1k8 on the Pi, 10k on the radio board.
P200-36 USB_VREG: radio-board output, never drive.

Radio supply

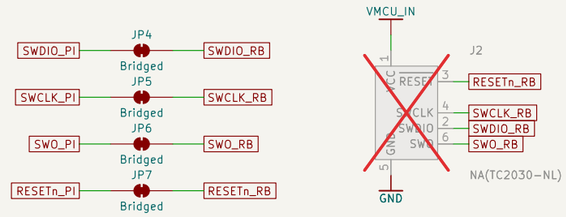
Pi 5 V -> FB1 -> LP5907 -> 3V3_RADIO -> R1 0R -> VMCU_IN (P201-2)



FB1 + C1/C2 keep the Pi 5 V noise out of the LDO. LP5907: 250 mA, 6.5 uVrms, EN tied to IN.
R1 0R = VMCU_IN current probe point. JP1 cut = board-ID EEPROM unpowered. R2 not fitted.

SWD cut points

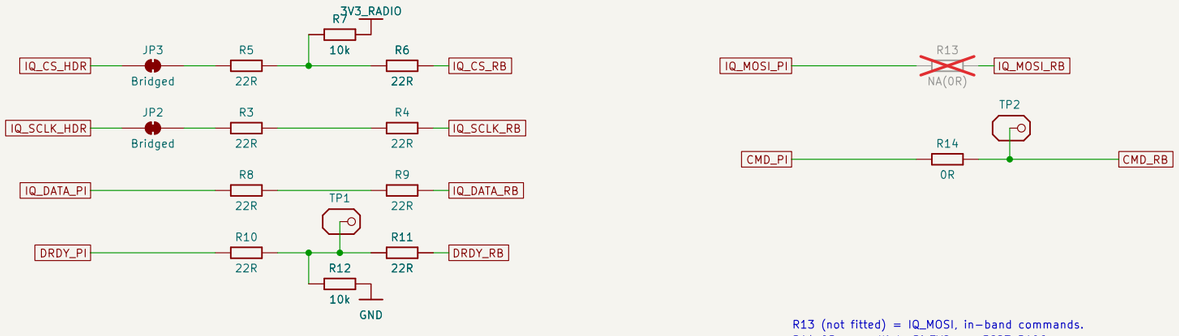
OpenOCD / pyOCD linuxgpiod from the Pi



Cut JP4-JP7 to use an external probe on J2
(Tag-Connect TC2030-NL, not fitted). RESETn: open-drain only.

I/Q link

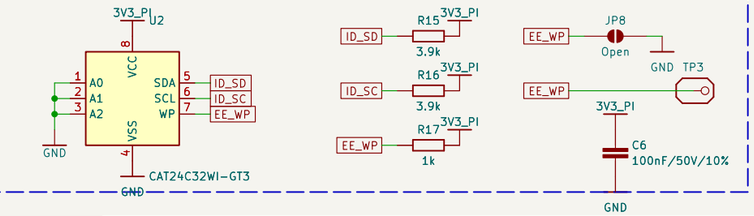
22 R at both ends (mode A: Pi drives SCLK/CS; mode B: FG23 drives them). JP2/JP3 = README SJ1/SJ2.



R13 (not fitted) = IQ_MOSI, in-band commands.
R14 0R = cmdlink, Pi TXD -> FG23 PA06.

ID EEPROM

RPi HAT design guide: CAT24C32 @ 0x50 on ID_SD / ID_SC, auto-loads the overlays



Generated by hardware/gen/gen_sch.py from design.py
github.com/ionos-sdr/ionos-maximus-rpi-sdr — CERN-OHL-P-2.0
Z2 Labs

Sheet: /
File: ionos_maximus.kicad_sch

Title: ionos-maximus RPi adapter (FG23 radio board)

Size: A3	Date: 2026-10-26	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 1/1